LESSON 7.4

# Compare and Order Integers and Rational Numbers

**Start here** · Words, worked example, and sentence starters

**Name** _____ **Date** _____ **Period** _____

**LEARNING TARGET** I can compare and order integers using a number line.

**WHY THIS MATTERS** Lucy and Yuzuki are playing a trivia game with these rules: if a player answers a question correctly, they receive 1 point; if they answer incorrectly, they lose 0.25 point. The player with the greatest score wins the game. The final scores for two games are shown in the table. Who is the winner of each game?

START HERE

## Words for this lesson

Read each word before you begin. Say it out loud.

| WORD | WHAT IT MEANS | EXAMPLE |
|---|---|---|
| **Number line** *Spanish: Recta numérica* | A straight line where numbers are placed in order; numbers get smaller to the left and larger to the right. | left is less, right is more |
| **Compare** *Spanish: Comparar* | To see if a number is bigger, smaller, or equal to another. | −7 compared with −3 |
| **Order** *Spanish: Ordenar* | To arrange numbers by size, from least to greatest or greatest to least. | Least to greatest: -5, -2, 0, 3, 7 (left to right on the number line) |
| **Greater than** *Spanish: Mayor que* | The > sign, showing one number is bigger. | −3 > −7 |
| **Less than** *Spanish: Menor que* | The < sign, showing one number is smaller. | −7 < −3 |
| **Negative integer** *Spanish: Entero negativo* | A whole number smaller than zero, like -5. | −7, −3, −100 |

HOW IT WORKS

## Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

**Compare −14 degrees and −5 degrees.**

1. Plot −14 and −5 on a number line.
2. −5 is to the right of −14.
3. Write the comparison: −14 < −5.

**Answer:** −5 is greater; −14 is colder

NOW YOU TRY

## Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

**Now let's compare -2 and 3. Which one is to the left on the number line? The -2 is left of zero.**

1. _____
2. _____
3. _____

**Answer:** _____

SAY IT AND WRITE IT

## Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- On the number line, _____ sits to the _____ of _____ , so _____ is greater.
- I can write the comparison as _____ _____ _____ .
- The least number in this set is _____ because _____ .
- Comparing two negative numbers is different from counting because _____ .

**Word bank** compare | order | greater than | less than | number line | left | right | negative | positive | least | greatest | inequality

WATCH OUT!

## A common mistake

A common mistake in Compare and Order Integers is thinking that a negative number with a bigger digit is the bigger value

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**Level 0** · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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**WHY THIS MATTERS** Lucy and Yuzuki are playing a trivia game with these rules: if a player answers a question correctly, they receive 1 point; if they answer incorrectly, they lose 0.25 point. The player with the greatest score wins the game. The final scores for two games are shown in the table. Who is the winner of each game?

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## PART 1 Understand the idea

models and meaning

**1** **Circle the letter of the best answer. Show how you know.**

Which statement is TRUE?

(A) -5 < -2

(B) -4 > 0

(C) -3 > -1

(D) -1 < -6

STEPS TO TRY

1. Sketch a quick number line. For negatives, 'greater' means closer to zero, not a bigger digit.
2. For each statement, place both numbers on the number line: the one farther LEFT is less. Test all four.
3. First statement: put -3 and -1 on the line. -3 is farther left, so is '-3 > -1' true or false? Keep testing.

SHOW YOUR WORK

**2** **Write the name of the correct group on each line.**

Drag to order these integers from LEAST to GREATEST.

**Groups:**

_____ 3

_____ -1

_____ -6

_____ 0

_____ -4

_____ 5

**STEPS TO TRY**

1. Least to greatest reads left to right on a number line — all the negatives come before 0.
2. Group first: negatives (-1, -6, -4), then 0, then positives (3, 5). Order the negatives by distance below zero.
3. Among -1, -4, and -6, the one farthest below zero goes first. Place it, then order the other two negatives.

## PART 2 Apply it

real-world problems

**3** **Circle the letter of the best answer. Show how you know.**

Which integer is the LEAST: -3, 5, -8, 2, 0?

(A) -8

(B) -3

(C) 0

(D) 5

**STEPS TO TRY**

1. 'Least' means farthest LEFT on the number line — for negatives, that's the one farthest below zero.
2. Place all five numbers on a mental number line. Negatives sit left of 0, and the least is the leftmost one.
3. Compare just the negatives first, -3 and -8: which is farther below zero? Then check it against 0, 2, and 5.

**SHOW YOUR WORK**

**4** **Circle the letter of the best answer. Show how you know.**

What is |−8|?

(A) 0
(B) 8
(C) −8
(D) −16

**STEPS TO TRY**

1. The bars |...| mean absolute value: the distance from 0, which is never negative.
2. Find -8 on the number line and count the units between it and 0.
3. Start at -8 and hop toward 0, counting each hop. The total number of hops is the value.

**SHOW YOUR WORK**

**EXPLAIN YOUR THINKING**

## How does plotting on the number line help you compare and order integers?

**SENTENCE STARTER** On the number line, ___ is to the left of ___, so ___ < ___. The order from least to greatest is ___.

**3-Point Exit Checklist:** ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

**How did it go?** ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help